Fixed-difference Kummer collector

KUMMER-DIFFERENTIAL-COLLECTOR-20261022

Completed targets per arm: 1260.

Uncached: 1260 source-verified, 230,430,039 quotient steps.

Uncached: 109,480,302 signed splits, 739 Recover failures.

Cold setup: 21 unique model builds, charged in all 1260 targets.

Source transfers and source verifications: 1260 each.

Every candidate and rho answer was independently checked as k·P=Q

Uncached cold ratio: 8.9597–55.9354; wins/losses: 0/36 cells.

on the original elliptic curve over Fp^2.

Model setup includes root search, theta search and torsion matrices.

Each cold target includes source transfer, initialization, the

quotient walk, signed Recover, graph elimination and verification.

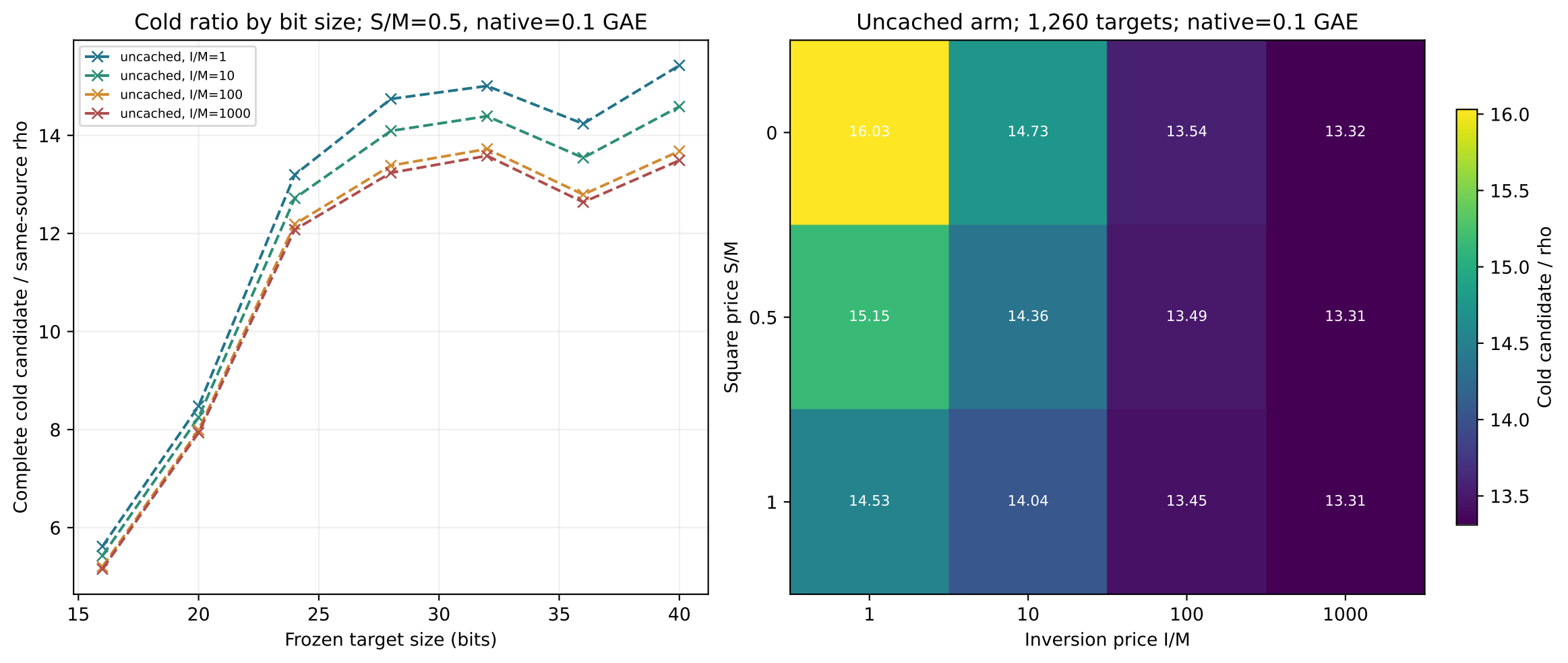
Raw: results/<arm>/collector-*.jsonl, cold-*-setup.jsonl,

results/target-price.csv and results/price-grid.csv.

All 36 cells use S/M in {0,0.5,1}, I/M in {1,10,100,1000},

and native cost in {0,0.1,1} GAE per declared logical operation.

Ratios describe this measured implementation and frozen cohort.


Cold ratio by bit size; S/M=0.5, native=0.1 GAE
uncached, I/M=1
uncached, I/M=10
uncached, I/M=100
uncached, I/M=1000
Complete cold candidate / same-source rho
Frozen target size (bits)
Uncached arm; 1,260 targets; native=0.1 GAE
Square price S/M
Inversion price I/M
16.03
14.73
13.54
13.32
15.15
14.36
13.49
13.31
14.53
14.04
13.45
13.31
Cold candidate / rho

uncached: native=0 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	9,798,151,036.8	175,168,978.8	55.935	0/1260
0.0	10	8,378,420,339.1	174,640,420.7	47.975	0/1260
0.0	100	7,075,915,111.8	174,155,505.1	40.630	0/1260
0.0	1000	6,830,646,535.8	174,064,192.9	39.242	0/1260
0.5	1	8,841,142,700.8	175,041,121.6	50.509	0/1260
0.5	10	7,966,770,880.2	174,617,698.3	45.624	0/1260
0.5	100	7,021,504,047.1	174,159,943.5	40.316	0/1260
0.5	1000	6,825,157,820.6	174,064,860.9	39.210	0/1260
1.0	1	8,157,565,318.0	174,949,795.0	46.628	0/1260
1.0	10	7,626,712,631.6	174,598,927.7	43.681	0/1260
1.0	100	6,969,019,037.8	174,164,224.7	40.014	0/1260
1.0	1000	6,819,690,778.5	174,065,526.2	39.179	0/1260

uncached: native=0.1 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	17,462,948,021.6	1,089,353,650.8	16.031	0/1260
0.0	10	16,043,217,323.9	1,088,825,092.7	14.734	0/1260
0.0	100	14,740,712,096.6	1,088,340,177.1	13.544	0/1260
0.0	1000	14,495,443,520.6	1,088,248,864.9	13.320	0/1260
0.5	1	16,505,939,685.6	1,089,225,793.6	15.154	0/1260
0.5	10	15,631,567,865.0	1,088,802,370.3	14.357	0/1260
0.5	100	14,686,301,031.9	1,088,344,615.5	13.494	0/1260
0.5	1000	14,489,954,805.4	1,088,249,532.9	13.315	0/1260
1.0	1	15,822,362,302.8	1,089,134,467.0	14.527	0/1260
1.0	10	15,291,509,616.4	1,088,783,599.7	14.045	0/1260
1.0	100	14,633,816,022.6	1,088,348,896.7	13.446	0/1260
1.0	1000	14,484,487,763.3	1,088,250,198.2	13.310	0/1260

uncached: native=1 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	86,446,120,884.8	9,317,015,698.8	9.278	0/1260
0.0	10	85,026,390,187.1	9,316,487,140.7	9.126	0/1260
0.0	100	83,723,884,959.8	9,316,002,225.1	8.987	0/1260
0.0	1000	83,478,616,383.8	9,315,910,912.9	8.961	0/1260
0.5	1	85,489,112,548.8	9,316,887,841.6	9.176	0/1260
0.5	10	84,614,740,728.2	9,316,464,418.3	9.082	0/1260
0.5	100	83,669,473,895.1	9,316,006,663.5	8.981	0/1260
0.5	1000	83,473,127,668.6	9,315,911,580.9	8.960	0/1260
1.0	1	84,805,535,166.0	9,316,796,515.0	9.102	0/1260
1.0	10	84,274,682,479.6	9,316,445,647.7	9.046	0/1260
1.0	100	83,616,988,885.8	9,316,010,944.7	8.976	0/1260
1.0	1000	83,467,660,626.5	9,315,912,246.2	8.960	0/1260